Chapter Review

Key Terms

complex fraction A complex fraction is a fraction in which the numerator or the denominator contains a fraction.

equivalent fractions Equivalent fractions are two or more fractions that have the same value.

fraction A fraction is written $\frac{a}{b}$. In a fraction, a is the numerator and b is the denominator. A fraction represents parts of a whole. The denominator b is the number of equal parts the whole has been divided into, and the numerator a indicates how many parts are included.

least common denominator (LCD) The least common denominator (LCD) of two fractions is the least common multiple (LCM) of their denominators.

mixed number A mixed number consists of a whole number a and a fraction $\frac{b}{c}$ where $c \neq 0$. It is written as $a\frac{b}{c}$, where $c \neq 0$.

proper and improper fractions The fraction $\frac{a}{b}$ is *proper* if $a < b$ and *improper* if $a \geq b$.

reciprocal The reciprocal of the fraction $\frac{a}{b}$ is $\frac{b}{a}$ where $a \neq 0$ and $b \neq 0$.

simplified fraction A fraction is considered simplified if there are no common factors in the numerator and denominator.

Key Concepts

4.1 Visualize Fractions

- **Property of One**
  - Any number, except zero, divided by itself is one.
    $\frac{a}{a} = 1$, where $a \neq 0$.
- **Mixed Numbers**
  - A **mixed number** consists of a whole number a and a fraction $\frac{b}{c}$ where $c \neq 0$.
  - It is written as follows: $a\frac{b}{c}$ $\quad$ $c \neq 0$
- **Proper and Improper Fractions**
  - The fraction $\frac{a}{b}$ is a proper fraction if $a < b$ and an improper fraction if $a \geq b$.
- **Convert an improper fraction to a mixed number.**
  Step 1. Divide the denominator into the numerator.
  Step 2. Identify the quotient, remainder, and divisor.
  Step 3. Write the mixed number as quotient $\frac{\text{remainder}}{\text{divisor}}$.
- **Convert a mixed number to an improper fraction.**
  Step 1. Multiply the whole number by the denominator.
  Step 2. Add the numerator to the product found in Step 1.
  Step 3. Write the final sum over the original denominator.
- **Equivalent Fractions Property**
  - If a, b, and c are numbers where $b \neq 0$, $c \neq 0$, then $\frac{a}{b} = \frac{a \cdot c}{b \cdot c}$.

4.2 Multiply and Divide Fractions

- **Equivalent Fractions Property**
  - If a, b, c are numbers where $b \neq 0$, $c \neq 0$, then $\frac{a}{b} = \frac{a \cdot c}{b \cdot c}$ and $\frac{a \cdot c}{b \cdot c} = \frac{a}{b}$.
- **Simplify a fraction.**
  Step 1. Rewrite the numerator and denominator to show the common factors. If needed, factor the numerator and denominator into prime numbers.
  Step 2. Simplify, using the equivalent fractions property, by removing common factors.
  Step 3. Multiply any remaining factors.
- **Fraction Multiplication**
  - If a, b, c, and d are numbers where $b \neq 0$and $d \neq 0$, then $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$.
- **Reciprocal**
  - A number and its reciprocal have a product of 1. $\frac{a}{b} \cdot \frac{b}{a} = 1$

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Opposite	Absolute Value	Reciprocal
has opposite sign	is never negative	has same sign, fraction inverts

Table 4.4

- **Fraction Division**
  - If a, b, c, and d are numbers where $b \neq 0$, $c \neq 0$ and $d \neq 0$, then
    $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$
    4.1
  - To divide fractions, multiply the first fraction by the reciprocal of the second.

4.3 Multiply and Divide Mixed Numbers and Complex Fractions

- **Multiply or divide mixed numbers.**
  Step 1. Convert the mixed numbers to improper fractions.
  Step 2. Follow the rules for fraction multiplication or division.
  Step 3. Simplify if possible.
- **Simplify a complex fraction.**
  Step 1. Rewrite the complex fraction as a division problem.
  Step 2. Follow the rules for dividing fractions.
  Step 3. Simplify if possible.
- **Placement of negative sign in a fraction.**
  - For any positive numbers a and b, $\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$.
- **Simplify an expression with a fraction bar.**
  Step 1. Simplify the numerator.
  Step 2. Simplify the denominator.
  Step 3. Simplify the fraction.

4.4 Add and Subtract Fractions with Common Denominators

- **Fraction Addition**
  - If a, b, and c are numbers where $c \neq 0$, then $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$.
  - To add fractions, add the numerators and place the sum over the common denominator.
- **Fraction Subtraction**
  - If a, b, and c are numbers where $c \neq 0$, then $\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$.
  - To subtract fractions, subtract the numerators and place the difference over the common denominator.

4.5 Add and Subtract Fractions with Different Denominators

- **Find the least common denominator (LCD) of two fractions.**
  Step 1. Factor each denominator into its primes.
  Step 2. List the primes, matching primes in columns when possible.
  Step 3. Bring down the columns.
  Step 4. Multiply the factors. The product is the LCM of the denominators.
  Step 5. The LCM of the denominators is the LCD of the fractions.
- **Equivalent Fractions Property**
  - If a, b, and c are whole numbers where $b \neq 0$, $c \neq 0$ then
    $\frac{a}{b} = \frac{a \cdot c}{b \cdot c}$ and $\frac{a \cdot c}{b \cdot c} = \frac{a}{b}$
- **Convert two fractions to equivalent fractions with their LCD as the common denominator.**
  Step 1. Find the LCD.
  Step 2. For each fraction, determine the number needed to multiply the denominator to get the LCD.
  Step 3. Use the Equivalent Fractions Property to multiply the numerator and denominator by the number from Step 2.
  Step 4. Simplify the numerator and denominator.
- **Add or subtract fractions with different denominators.**
  Step 1. Find the LCD.

Step 2. Convert each fraction to an equivalent form with the LCD as the denominator.
Step 3. Add or subtract the fractions.
Step 4. Write the result in simplified form.

- **Summary of Fraction Operations**
  - **Fraction multiplication:** Multiply the numerators and multiply the denominators.
    $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$
    4.2
  - **Fraction division:** Multiply the first fraction by the reciprocal of the second.
    $\frac{a}{b} + \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$
    4.3
  - **Fraction addition:** Add the numerators and place the sum over the common denominator. If the fractions have different denominators, first convert them to equivalent forms with the LCD.
    $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$
    4.4
  - **Fraction subtraction:** Subtract the numerators and place the difference over the common denominator. If the fractions have different denominators, first convert them to equivalent forms with the LCD.
    $\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}$
    4.5

- **Simplify complex fractions.**
  Step 1. Simplify the numerator.
  Step 2. Simplify the denominator.
  Step 3. Divide the numerator by the denominator.
  Step 4. Simplify if possible.

4.6 Add and Subtract Mixed Numbers

- **Add mixed numbers with a common denominator.**
  Step 1. Add the whole numbers.
  Step 2. Add the fractions.
  Step 3. Simplify, if possible.

- **Subtract mixed numbers with common denominators.**
  Step 1. Rewrite the problem in vertical form.
  Step 2. Compare the two fractions.
  If the top fraction is larger than the bottom fraction, go to Step 3.
  If not, in the top mixed number, take one whole and add it to the fraction part, making a mixed number with an improper fraction.
  Step 3. Subtract the fractions.
  Step 4. Subtract the whole numbers.
  Step 5. Simplify, if possible.

- **Subtract mixed numbers with common denominators as improper fractions.**
  Step 1. Rewrite the mixed numbers as improper fractions.
  Step 2. Subtract the numerators.
  Step 3. Write the answer as a mixed number, simplifying the fraction part, if possible.

4.7 Solve Equations with Fractions

- **Determine whether a number is a solution to an equation.**
  Step 1. Substitute the number for the variable in the equation.
  Step 2. Simplify the expressions on both sides of the equation.
  Step 3. Determine whether the resulting equation is true. If it is true, the number is a solution. If it is not true, the number is not a solution.

- **Addition, Subtraction, and Division Properties of Equality**
  - For any numbers a, b, and c,
    if $a = b$, then $a + c = b + c$. Addition Property of Equality
  - if $a = b$, then $a - c = b - c$. Subtraction Property of Equality